Reliability Beyond Marginal Coverage for Observed Compositions

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Abstract

Conformal regions for an observed composition require simultaneous-vector coverage, subgroup diagnostics and a common measure of efficiency. We compare split-conformal scores and calibration schemes in controlled logistic-normal regression using 100 independent full-dataset repetitions per main cell, three or ten components, and a shared fitted predictor. At three components with heterogeneous noise and 499 calibration observations, global Aitchison joint coverage is $0.9017[0.8988, 0.9045]$, but high-noise-group coverage is only $0.5069[0.4931, 0.5207]$. A shift increasing that group's frequency reduces marginal coverage to $0.6050[0.5944, 0.6157]$, while observed-group calibration reaches $0.9055[0.9013, 0.9097]$. Learned scaling improves adaptation but still misses the group target in a ten-component case; correct oracle weighting restores shifted marginal coverage without guaranteeing group coverage. Common Euclidean simplex-area estimates reveal setting-dependent efficiency differences, with coverage reported alongside size. Coordinatewise calibration can substantially undercover the whole vector, and zero replacement requires distinguishing direct raw-response membership from an explicit inverse-image region, which inherits transformed coverage for original responses. The contribution is a reproducible diagnostic evaluation under explicit synthetic mechanisms, not a new coverage theorem or a universal method ranking.

1 Introduction

A predicted probability vector and an uncertain compositional response are different statistical objects. A classifier may output a vector of probabilities while its observed outcome is a single label. Here the response itself is observed as a nonnegative vector summing to one. Examples include measured proportions and repeated-measurement distributions. Our coverage event is simultaneous inclusion of this entire vector in a region of the simplex; covering a label or each coordinate separately answers a different question.

Split conformal prediction provides a finite-sample marginal coverage guarantee under exchangeability, for a score fixed independently of calibration data. This leaves several choices open. Raw-coordinate and log-ratio distances produce different regions. A global calibration threshold averages over heterogeneous populations. Smaller regions can reflect undercoverage, and radii in different geometries have incompatible units. Finally, a guarantee averaging over calibration samples need not describe the performance of every realized calibration sample.

We study these choices in a controlled, synthetic compositional regression benchmark. The design uses a shared fitted predictor, independent repetitions of the full training–calibration–test experiment, and explicit subgroup and distribution-shift diagnostics. It separates four questions: whether coordinatewise calibration delivers simultaneous coverage; whether marginal coverage conceals

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subgroup failures; how geometry changes efficiency under a common reference measure; and which conclusions survive a controlled mixture shift, finite calibration ranks, and zero replacement. The contribution is an evaluation design and its reproducible evidence, rather than a new conformal guarantee. The controlled setting exposes mechanisms; it cannot establish performance for a particular application or a universal ranking of methods.

Our theory-based expectations are that correctly ranked joint scores achieve iid marginal coverage, naive coordinate calibration can miss the joint target, and heterogeneity can hide group undercoverage that becomes more consequential under mixture shift. We test whether scaling or group calibration improves this behavior and changes common-measure area; no geometry is presumed universally best. These hypotheses concern the controlled mechanisms, not all compositions.

2 Related work and positioning

Split conformal prediction separates fitting from calibration and gives marginal coverage under exchangeability; its computational simplicity motivates holding the predictor fixed while varying scores and calibration schemes [Lei et al., 2017, Angelopoulos and Bates, 2022]. Full conformal and refitting-based alternatives make different computational and statistical tradeoffs. Jackknife+ uses leave-one-out fits and has a general worst-case $1 - 2\alpha$ lower bound in its ordinary construction [Barber et al., 2020]. We do not benchmark every conformal family or claim a new rank guarantee.

The closest permitted work is Javanmardi et al. [2024]. They already split-conformalize distribution-vector labels into simplex regions, compare scores including total variation, KL divergence, Wasserstein distance and model-based alternatives, and evaluate coverage and relative simplex size across repeated splits. Thus conformalizing a composition, varying distributional scores and measuring a fraction of the simplex are established ideas. Their distinction between noisy observed distribution labels and latent clean probabilities is also directly relevant: coverage for one does not automatically establish coverage for the other. Our focus is a shared clr-regression predictor under controlled heterogeneity, coordinate simultaneity, calibration variation, a known group-mixture intervention and boundary sensitivity. The contribution must be assessed through those completed comparisons, rather than through a claim that simplex regions are new.

Classification prediction sets instead cover labels, even when their scores use a softmax vector [Angelopoulos et al., 2022]. Class-conditional and clustered calibration illustrate how marginal guarantees can hide class failures and how fine strata increase calibration demands [Ding et al., 2023]. Our strata are observed features, not response classes. Multivalid conformal methods address richer collections of groups and thresholds [Jung et al., 2022]; evaluating two groups or calibrating within them does not establish such a general property.

Weighted conformal prediction handles specified covariate shift when the conditional response law is invariant and appropriate likelihood ratios are known [Tibshirani et al., 2020]. The oracle baseline here isolates the effect of exact ratios under a deliberately transparent shift. It is not evidence that estimated weights work without additional assumptions. Finally, coverage parity does not itself establish equitable downstream decisions: human-subject classification evidence makes this distinction concrete [Cresswell et al., 2025]. No human-utility or fairness conclusion follows from this numerical benchmark.

3 Prediction target and calibration

Let $\Delta^{K-1} = \{y \in [0,1]^K : \sum_j y_j = 1\}$, with interior Δ_+^{K-1}. We observe independent training and calibration samples of pairs (X, Y) and predict a fresh composition Y. A fitted center $\hat{p}(x) \in \Delta_+^{K-1}$ and positive scale $\hat{s}(x)$ use training data only. For a scalar nonconformity score S, let

$$k = \lceil (n+1)(1-\alpha) \rceil, \qquad q = S_{(k)}, \qquad C(x) = \{y : S(x,y) \le q\}, \tag{1}$$

where $S_{(k)}$ is the kth smallest calibration score and $q = +\infty$ if $k > n$. Finite regions are restricted to their stated simplex domain; the implementation explicitly defines an infinite threshold as a whole-simplex fallback, including boundary points. Conditional on training data, exchangeable calibration and test pairs give $\Pr\{Y \in C(X)\} \ge 1 - \alpha$. With no score ties the probability is $k/(n+1)$, capped at one. This probability averages over calibration data and the new pair. It does not assert coverage conditional on a realized calibration set or at each x.

Scores and controls. Raw L2 uses $\|y - \hat{p}(x)\|_2$; raw max uses $\|y - \hat{p}(x)\|_\infty$. For positive compositions, $\mathrm{clr}(y)_j = \log y_j - K^{-1} \sum_\ell \log y_\ell$. Aitchison uses $\| \mathrm{clr}(y) - \mathrm{clr}(\hat{p}(x))\|_2$. Scaled Aitchison divides this score by the training-fitted $\hat{s}(x)$. These are joint scores: one calibration rank controls inclusion of the whole composition. Orthonormal ilr coordinates give exactly the same norm and region as clr, so they are not a separate baseline.

Mondrian Aitchison uses a separately ranked threshold q_g within each predeclared, observed group $G = g$. Conditional on the group calibration count, the same rank argument applies within that group when its calibration and test laws agree; an insufficient count returns an unbounded threshold. Learned scaling alone has no analogous group guarantee. Two coordinatewise controls calibrate $|Y_j - \hat{p}_j(X)|$ separately: naive calibration uses error α per coordinate, and Bonferroni uses α/K. Their regions intersect all coordinate constraints with the simplex. Only the latter receives the union-bound joint guarantee. Simplex dependence does not generally turn the naive control into a joint-$1 - \alpha$ method.

Prior-work score supplements. Matched-seed supplements use total variation $\frac{1}{2} \sum_j |y_j - \hat{p}_j(x)|$ and forward KL $\sum_j y_j \log(y_j/\hat{p}_j(x))$, with the same split ranking. Forward KL is asymmetric and is not a metric; zero response terms, when evaluated, use $0 \log 0 = 0$ with strictly positive predicted components. At $K = 3$, the sum-zero residual has one sign represented by a single component, so half its L1 norm equals its maximum absolute component. Total variation therefore duplicates raw max in this dimension and cannot count as independent evidence. It differs at $K = 10$.

Known-ratio shift reference. An oracle weighted Aitchison baseline uses the exact covariate density ratio of the synthetic shift, $w(X) = 4$ for $G = 1$ and $w(X) = 0.25$ for $G = 0$, with unit weights in the iid environment. At candidate feature x, the threshold is the $(1 - \alpha)$ quantile of

$$\frac{\sum_{i=1}^n w(X_i)\delta_{S_i} + w(x)\delta_{+\infty}}{\sum_{i=1}^n w(X_i) + w(x)}. \tag{2}$$

The test-point mass at infinity is part of the construction. This baseline uses generator knowledge; it is not an estimated-ratio deployment method. Its shifted marginal guarantee needs the stated covariate shift and correct ratios, and does not give arbitrary subgroup coverage.

Geometry and efficiency. Coverage uses direct region membership. At $K = 3$, efficiency is the expected fraction of Euclidean simplex area occupied by the region, estimated using independent feature anchors and common uniform-simplex candidates across methods. This makes comparisons between raw and clr regions meaningful under one declared measure. It does not make Euclidean area an application-independent utility. Finite radii may be reported as within-geometry diagnostics, with infinite thresholds retained explicitly; radii are never ranked across raw and log-ratio units. We do not estimate cross-geometry volume at $K = 10$.

4 Experimental design

Controlled response model. Features are $X = (G, Z_1, Z_2, Z_3)$, where $G \sim \mathrm{Bernoulli}(0.2)$ and the Z_j are independent standard normals, independent of G. With $\phi(X) = (1, G, Z_1, Z_2, Z_3, Z_1Z_2, Z_1^2)$ and a fixed, dimension-specific coefficient matrix B, generate

$$Y = \mathrm{softmax}\{\phi(X)B + \sigma(X)(a \odot E)\}, \quad E \sim N(0, I_K). \tag{3}$$

The positive axis weights a increase linearly from 0.65 to 1.35 before normalization to mean square one. One coefficient matrix per dimension is fixed across replications; the study does not average over coefficient draws. Homogeneous noise has $\sigma(X) = 0.45$. Heterogeneous noise has $\sigma(X) = 0.18 \exp(0.15Z_1)$ for $G = 0$ and $0.95 \exp(0.15Z_1)$ for $G = 1$. This deliberately creates a known high-noise minority group. All constants and coefficient seeds are fixed in the executable generator.

The predictor fits ridge regression of clr responses on ϕ, using 1,000 independent training observations and penalty 10^{-3} on every coefficient except the intercept. Its predicted composition is the softmax of fitted logits. A second training-only ridge regression predicts log squared clr-residual norm, whose clipped half-exponential gives the positive scale (Appendix B). The predictor estimates a clr center; it is not generally the conditional arithmetic mean of Y. The correctly specified log-ratio mean structure favors this modeling approach and limits extrapolation to misspecified predictors.

Matrix and shift. The prespecified main matrix crosses $K \in \{3, 10\}$, calibration size $n \in \{99, 499\}$ and homogeneous/heterogeneous noise, at $\alpha = 0.1$. Each cell uses 100 fresh training/calibration/test repetitions, separate from implementation-pilot seeds, and 5,000 test observations per environment. Methods share observations within a repetition. The shifted environment changes only $\Pr(G = 1)$ from 0.2 to 0.8, preserving the joint feature/response law within each group. It therefore challenges global calibration while preserving the assumptions of observed-group calibration. It does not represent arbitrary covariate or concept shift.

Design chronology. The seven core scores/calibration controls were fixed after an implementation/runtime pilot and before the main run. The known-ratio reference was proposed before confirmatory outcomes. The plan owner requested TV and forward-KL after the core job started for relevance to newly supplied permitted prior work, without inspecting core outcomes. Analysis had inspected core/oracle outcomes when freezing and implementing that supplement, but had not inspected TV/KL outcomes. These are source-driven matched-seed additions, not original-core prespecification or a fully blind extension; all original core results are retained.

Focused scope checks. A separate rank experiment varies calibration size and nominal error with continuous and tied scores, including ranks that require an infinite threshold. A boundary sensitivity censors small components to exact zeros and compares original-target raw calibration with consistently transformed clr calibration on $T_\epsilon(Y)$. It uses 50 fresh repetitions per dimension, $n = 499$, the heterogeneous mechanism and $\epsilon \in \{10^{-8}, 10^{-4}, 10^{-2}\}$. The same original data are shared across replacements. A separately labeled diagnostic applies transformed-calibrated clr regions to original responses; it has no original-target nominal guarantee. An explicit inverse-image region instead inherits transformed coverage for original responses by exact event equivalence. Appendix B specifies both membership conventions.

Uncertainty and reporting. The unit of uncertainty is an independently generated full dataset repetition, not an individual test observation. We report pointwise 95% Student-t intervals for mean metrics and paired differences between methods using the same repetition. These intervals include training, calibration and finite-test variability; they are not simultaneous intervals over all comparisons. Test-estimated coverage distributions additionally describe realized-calibration variation, without certifying population coverage for individual calibration sets. Efficiency estimates use separate feature anchors; candidate Monte Carlo precision is retained separately from across-repetition variation. The main area estimate uses 32 new feature anchors and 8,192 candidates per anchor and environment. Unresolved tiny areas do not establish zero volume or a reliable ranking.

5 Results

Unless specified otherwise, numerical examples below use the heterogeneous mechanism, $n = 499$, and pointwise 95% intervals across independent full-dataset repetitions. They illustrate prespecified comparisons; all dimensions, calibration sizes, scenarios and environments remain in the delivered cell summaries. An interval near nominal is not an equivalence test or proof of exact validity.

Joint calibration is necessary for joint reliability. At $K = 3$, iid joint coverage is $0.8390[0.8357, 0.8424]$ for naive coordinate calibration, $0.9360[0.9339, 0.9380]$ for Bonferroni, $0.9016[0.8987, 0.9044]$ for raw L2 and $0.9017[0.8988, 0.9045]$ for global Aitchison. The naive control falls to $0.6380[0.6328, 0.6432]$ at $K = 10$. Thus coordinatewise calibration can fail substantially at the vector level, despite each coordinate being calibrated separately. Bonferroni is a valid but conservative control. Joint-score iid marginal means across the core matrix remain close to their rank targets; this does not mean every individual pointwise interval contains 0.9 or that regions have equal size.

Marginal success conceals subgroup failure. For global Aitchison at $K = 3$, low-noise-group coverage is $0.9999[0.9999, 1.0000]$ while high-noise-group coverage is $0.5069[0.4931, 0.5207]$. Their mixture produces acceptable marginal coverage while allocating misses very unevenly (Figure 1; Table 1). Mondrian calibration gives low/high group coverages $0.9019[0.8985, 0.9053]$ and $0.9063[0.9010, 0.9116]$; learned scaling gives $0.9013[0.8983, 0.9043]$ and $0.8988[0.8935, 0.9041]$.

Table 1: **Reliability and common area in the heterogeneous $K = 3$ setting.** Calibration size is 499; entries show replication means $\pm$ pointwise 95% interval half-widths, not standard deviations across runs. The shifted group-1 probability is 0.8 versus 0.2 for iid evaluation. Area is a fraction of Euclidean simplex area, averaged over iid feature anchors. Smaller area is useful only with adequate coverage; coordinatewise naive is the negative control. TV exactly duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. These intervals are not simultaneous. Full-cell estimates and replication counts are supplied in `tables/all_estimates.csv`.

Method	Joint iid	Group 1 iid	Joint shifted	Area fraction iid
Raw Euclidean	0.902 ± 0.003	0.512 ± 0.013	0.608 ± 0.010	0.232 ± 0.010
Raw maximum	0.902 ± 0.003	0.512 ± 0.013	0.609 ± 0.010	0.230 ± 0.010
Total variation	0.902 ± 0.003	0.512 ± 0.013	0.609 ± 0.010	0.230 ± 0.010
Forward KL	0.902 ± 0.003	0.511 ± 0.012	0.608 ± 0.010	0.216 ± 0.008
Aitchison	0.902 ± 0.003	0.507 ± 0.014	0.605 ± 0.011	0.253 ± 0.010
Scaled Aitchison	0.901 ± 0.003	0.899 ± 0.005	0.898 ± 0.004	0.157 ± 0.008
Group Aitchison	0.903 ± 0.003	0.906 ± 0.005	0.905 ± 0.004	0.164 ± 0.008
Oracle weighted	0.902 ± 0.003	0.507 ± 0.014	0.907 ± 0.005	0.253 ± 0.010
Coord. naive	0.839 ± 0.003	0.324 ± 0.007	0.451 ± 0.006	0.090 ± 0.003
Coord. Bonferroni	0.936 ± 0.002	0.681 ± 0.009	0.744 ± 0.008	0.317 ± 0.009

Both adapt strongly in this mechanism. The negative result matters: at $K = 10$, scaled high-noise-group coverage is $0.8940[0.8893, 0.8988]$, below nominal in this pointwise estimate. A learned scale is useful here without acquiring a distribution-free group guarantee.

Geometry and adaptation change measured area. At $K = 3$, mean iid Euclidean simplex-area fractions are $0.2525[0.2429, 0.2621]$ for global Aitchison, $0.2316[0.2218, 0.2414]$ for raw L2 and $0.2303[0.2206, 0.2400]$ for raw max. Scaling and Mondrian give $0.1574[0.1495, 0.1654]$ and $0.1636[0.1552, 0.1721]$. Their paired area differences from global Aitchison are $-0.0951[-0.1080, -0.0821]$ and $-0.0889[-0.1023, -0.0755]$. Figure 2 puts these areas beside the homogeneous comparison. These results support a smaller expected area for adaptive regions in this iid heterogeneous setting, with coverage shown alongside size. They do not make raw/global regions group-valid, establish a measure-independent ranking or transfer to $K = 10$. Some other prespecified area contrasts remain unresolved, including raw L2/max versus global Aitchison under heterogeneous shift at $n = 99$ and scaled versus global Aitchison under homogeneous shift. We withhold their orderings.

Prior-work scores do not remove the diagnostic failures. The source-driven controls give TV joint coverage $0.9017[0.8988, 0.9045]$ and forward-KL coverage $0.9022[0.8997, 0.9048]$ in the same $K = 3$ iid cell. Forward-KL high-noise-group coverage is $0.5110[0.4988, 0.5231]$ and shifted marginal coverage is $0.6083[0.5985, 0.6181]$. Thus changing these score geometries alone does not repair group or unweighted-shift failures. Forward-KL area is $0.2158[0.2075, 0.2240]$, with paired KL-minus-raw-L2 difference $-0.0158[-0.0197, -0.0120]$. This is a smaller iid area under the chosen reference, not a universal optimum. All 800 matched $K = 3$ TV coverage/group/area rows exactly duplicate raw max; TV is distinct at $K = 10$. Appendix Table 3 gives the paired area contrasts. These comparisons hold our predictor fixed rather than reproduce a prior study's complete fitted pipeline.

The controlled shift exposes the hidden misses. At $K = 3$, changing the high-noise-group frequency to 0.8 reduces global Aitchison marginal coverage to $0.6050[0.5944, 0.6157]$, a paired shifted-minus-iid difference of $-0.2966[-0.3046, -0.2886]$. The group-mixture identity in Appendix A explains the direction and scale of this failure. Mondrian reaches $0.9055[0.9013, 0.9097]$, oracle known-ratio weighting $0.9066[0.9019, 0.9113]$ and learned scaling $0.8976[0.8934, 0.9017]$. Oracle high-noise-group coverage is only $0.8831[0.8772, 0.8889]$: restoring target-marginal coverage does not restore group coverage even with correct ratios.

Mondrian and oracle weighted areas under shift are $0.4809[0.4679, 0.4938]$ and $0.5632[0.5514, 0.5749]$; the paired oracle-minus-Mondrian difference is $0.0823[0.0724, 0.0921]$. A small global region that severely undercovers is not a reliable efficiency winner. These comparisons

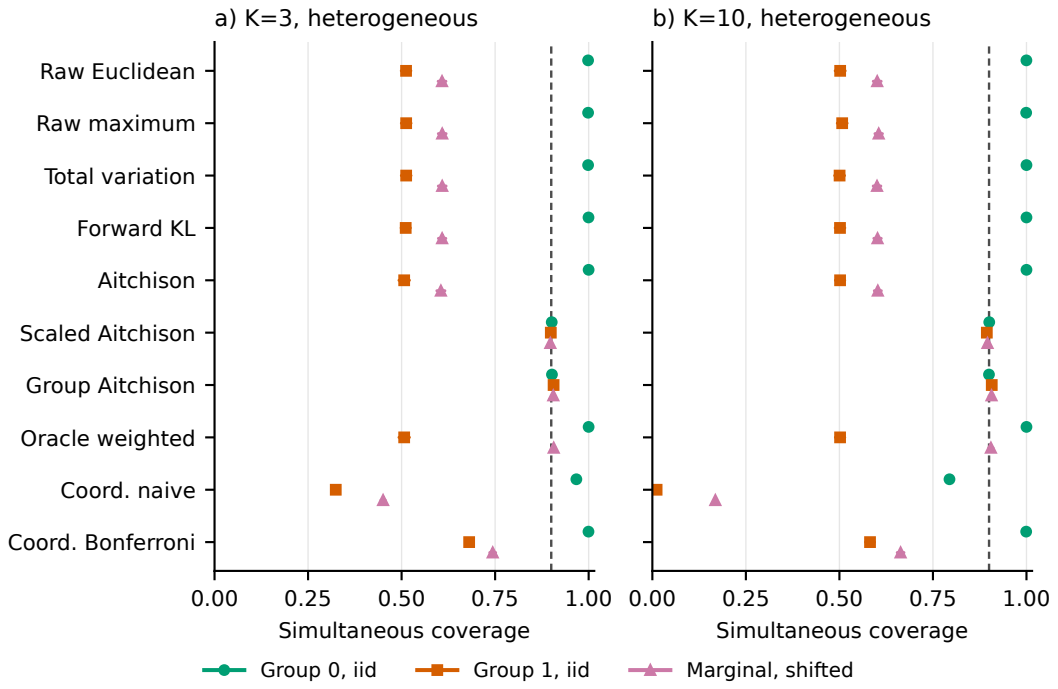


Figure 1: **Subgroup coverage and controlled mixture shift.** Dots are means across 100 independent train/calibration/test replications; bars are pointwise 95% t intervals for these means, not simultaneous bands or intervals for a single calibrated run. Each replication uses 1000 training and 499 calibration observations and 5000 test observations per environment. Circles and squares are iid coverage in the prespecified lower-noise group 0 and higher-noise group 1. Triangles are marginal coverage after the test group-1 probability changes from 0.2 to 0.8, with within-group laws preserved. Colors and shapes distinguish estimands; shifted marginal coverage is not a third subgroup. "Group Aitchison" calibrates within observed groups; "Oracle weighted" uses the known density ratio, unavailable in general applications. The dashed line marks 0.90. TV exactly duplicates raw maximum at $K = 3$; KL is forward $y\|\hat{\mu}$. This experiment does not test arbitrary shift.

exploit known groups or ratios and invariant within-group laws; they do not test unknown groups, learned ratios or arbitrary shift. Unit-weight oracle results coincide exactly with global Aitchison on all iid matched rows, verifying algorithm and stream matching rather than an independent replication of that method.

Mean coverage does not certify each calibration sample. For global Aitchison at $K = 3$, iid finite-test coverage has across-repetition standard deviation 0.0263 at $n = 99$ and 0.0145 at $n = 499$. The empirical fractions below 0.85 are $0.0200[0.0055, 0.0700]$ and $0.0000[0.0000, 0.0370]$, with Wilson intervals. Even zero observed failures has a positive upper confidence bound. These events concern finite-test estimates; they are not guarantees about population coverage conditional on every realized calibration set.

Zero replacement changes what is covered. At $K = 10$ and $\epsilon = 10^{-8}$, transformed-target global Aitchison coverage is $0.8990[0.8954, 0.9026]$. Testing the same transformed-calibrated region directly against original responses gives only $0.4721[0.4702, 0.4740]$, whereas separately calibrated original-target raw L2 gives $0.9000[0.8955, 0.9045]$. The boundary-response frequency is $0.5279[0.5260, 0.5298]$. For transformed global Aitchison, the paired 10^{-2}-minus-10^{-8} coverage change is $0.0008[-0.0020, 0.0036]$, an unresolved mean-coverage change despite changed targets and fitted geometry. The full epsilon grid quantifies sensitivity on paired original datasets; changing ϵ changes both fitted geometry and the transformed target. The direct-original diagnostic is a different region from the explicit inverse image $C_{\text{pre}}(x) = \{y : T_\epsilon(y) \in C_\epsilon(x)\}$. The latter has exactly the measured transformed membership event, so its original-response coverage and interval

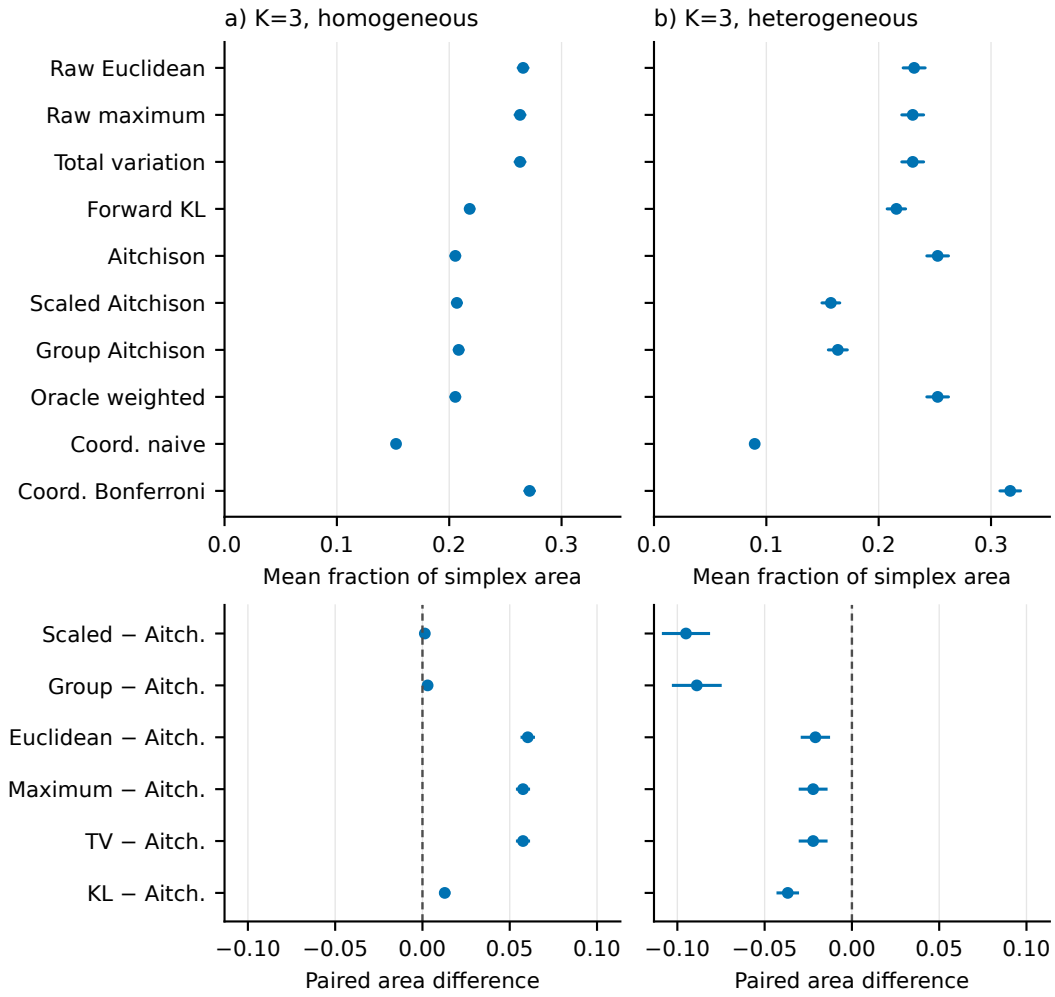


Figure 2: **A common area measure for $K = 3$.** Dots are means across 100 independent train/calibration/test replications; bars are pointwise 95% t intervals for these means, not simultaneous bands or intervals for a single calibrated run. Each replication uses 1000 training and 499 calibration observations and 5000 test observations per environment. Area uses 32 independent feature anchors and 8192 common uniform-simplex candidates per anchor in each replication/environment. Top: mean fraction of Euclidean simplex area under iid sampling. The uniform-simplex reference measure is shared across methods; it is not clr volume or application utility. Bottom: paired within-replication differences from unscaled Aitchison, with pointwise 95% t intervals; negative values indicate smaller area. Interpret area jointly with the coverage displays; the naive control has no nominal joint-coverage guarantee. TV exactly duplicates raw maximum at $K = 3$, so these rows are one geometry. KL is forward $y\|\hat{\mu}$. Intervals crossing zero leave area orderings unresolved. No cross-geometry size comparison is made at $K = 10$.

are also $0.8990[0.8954, 0.9026]$, including zeros. This is an identity from existing events, not a new simulation; its original-simplex area was not estimated. Censoring tests one zero mechanism.

Numerical checks and executed scope. The rank study executes all 30 declared continuous/tied-score cells, each with 20,000 trials, and compares empirical coverage/calibration variability with the exact rank and beta/discrete references. For uniform scores at $n = 99$, $\alpha = 0.1$, test coverage is $0.8976[0.8934, 0.9018]$ against theoretical 0.9; at $n = 9$ with tied Bernoulli scores it is $0.9991[0.9986, 0.9995]$. Infinite-threshold cases return full coverage, while ties are conservative. Separate checks passed for clr/ilr equivalence, perturbation invariance, analytic simplex area, Bonferroni nesting, empty/tiny groups, weighted test atoms and finite/infinite serialization. No core,

oracle or boundary repetition was excluded and no resource run failed. The resource-reported core, initial-extension and TV/KL running times were 116.110, 43.878 and 44.498 seconds, respectively, excluding startup; source hashes, timings, the pilot and the pre-main serialization correction are retained with the code. The source-to-claim map and inline-quantity extraction accompany the manuscript.

6 Discussion and limitations

The evaluation separates the target, the validity assumptions and the efficiency measure. A joint region must be judged by simultaneous vector coverage. Marginal validity must be accompanied by subgroup diagnostics when subgroup reliability matters. An efficiency comparison needs both coverage and a shared measure of size. These principles describe how to interpret the experiments; the evidence supports only the specific models, distributions and interventions actually tested.

Method choice under the tested conditions. Use a joint score or an explicit multiplicity correction when the whole vector must be covered; naive coordinate calibration failed that target. For predeclared observed groups requiring separate reliability, group calibration fits the tested within-group-preserving shift, subject to sparse-group costs. Training-only scaling improves adaptation and expected iid area here, but its measured high-noise-group miss at ten components rules out a general group guarantee. Known-ratio weighting addresses shifted marginal coverage under its assumptions, while the oracle group miss shows why the two goals remain distinct. For zeros, calibrate a boundary-compatible score on the original response or declare a transformed/inverse-image target explicitly. No single raw or log-ratio geometry is an application-independent efficiency winner.

The synthetic design offers known ground truth, controlled heterogeneity and repeatable independent calibration experiments. It lacks a real application, an application-specific loss for region size, and a diversity of fitted predictors. The generator's log-ratio mean belongs to the fitted basis. Observed groups are supplied without estimation error, and the shift preserves their conditional laws exactly. Unknown groups, misspecified density ratios, changes within groups and deployment-time dependence can invalidate the corresponding interpretation. Neither a learned scale nor good empirical subgroup coverage establishes distribution-free pointwise conditional coverage.

Euclidean simplex area is only one reference measure. It emphasizes different parts of the simplex from a log-ratio measure and is evaluated only at $K = 3$. Results at $K = 10$ concern coverage and declared within-geometry diagnostics. We do not infer a high-dimensional efficiency ranking from low-dimensional area. Pointwise intervals across many cells do not provide multiplicity-controlled discoveries or prove exact validity from finite simulation.

Ordinary clr is undefined at zeros. A positive replacement applied consistently to all splits produces a different, exchangeable response $T_\epsilon(Y)$; its coverage does not guarantee direct membership of the original response in the transformed clr region. An explicit inverse-image region $\{y : T_\epsilon(y) \in C_\epsilon(x)\}$ does inherit the same coverage for original compositions, including zeros, by exact event identity. Its original-simplex area was not evaluated. Censoring in the sensitivity study is a measurement mechanism, not a model for every kind of structural zero. Choosing a scientifically justified zero mechanism remains application dependent.

We compare split-conformal score and calibration choices with one shared predictor. Full conformal, jackknife+, learned density regions and other multivariate methods may trade computation, adaptation and efficiency differently. Their absence restricts the comparison; it does not imply inferiority. No novelty claim rests on exhaustive discovery of the literature, and no conclusion is claimed for latent class probabilities inferred from one observed label.

Potential impacts. Better diagnostics can prevent users from treating an average coverage guarantee as assurance for an undercovered population. Conversely, presenting a compact region or a nominal guarantee without its target, group and shift conditions can encourage unwarranted confidence in consequential decisions. This synthetic study neither measures downstream harm nor certifies equitable decisions. Deployment would require relevant data, subgroup definitions, scientific treatment of zeros and monitoring for changes beyond the tested mixture intervention.

References

Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification, 2022. URL https://arxiv.org/abs/2107.07511. Consulted version 6.

Anastasios N. Angelopoulos, Stephen Bates, Jitendra Malik, and Michael I. Jordan. Uncertainty sets for image classifiers using conformal prediction, 2022. URL https://arxiv.org/abs/2009.14193. Consulted version 5.

Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Predictive inference with the jackknife+, 2020. URL https://arxiv.org/abs/1905.02928. Consulted version 3.

Jesse C. Cresswell, Bhargava Kumar, Yi Sui, and Mouloud Belbahri. Conformal prediction sets can cause disparate impact, 2025. URL https://arxiv.org/abs/2410.01888. Consulted version 2.

Tiffany Ding, Anastasios N. Angelopoulos, Stephen Bates, Michael I. Jordan, and Ryan J. Tibshirani. Class-conditional conformal prediction with many classes, 2023. URL https://arxiv.org/abs/2306.09335. Consulted version 2.

Alireza Javanmardi, David Stutz, and Eyke Hüllermeier. Conformalized credal set predictors, 2024. URL https://arxiv.org/abs/2402.10723. Consulted version 1.

Christopher Jung, Georgy Noarov, Ramya Ramalingam, and Aaron Roth. Batch multivalid conformal prediction, 2022. URL https://arxiv.org/abs/2209.15145. Consulted version 1.

Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression, 2017. URL https://arxiv.org/abs/1604.04173. Consulted version 2.

Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. Conformal prediction under covariate shift, 2020. URL https://arxiv.org/abs/1904.06019. Consulted version 3.

A Rank, variability and shift calculations

These are standard calculations and implementation checks, not new distribution-free theorems. Condition on the training sample so that the score is fixed. With exchangeable continuous calibration and test scores, the test score has a uniform rank among $n+1$ scores. Inclusion in (1) is equivalent to rank at most k, proving $k/(n+1)$ coverage. Ties make the unrandomized comparison conservative. If $k > n$, the infinite threshold includes the score domain. Finite clr regions have the positive simplex as their domain. The implementation separately defines an infinite threshold as a whole-simplex fallback, including the boundary; this is an explicit rule, not evaluation of an undefined boundary clr score.

If scores are iid with continuous CDF F and $k \leq n$, the probability integral transform gives

$$F(q) \sim \mathrm{Beta}(k, n+1-k), \qquad \mathrm{Var}\{F(q)\} = \frac{k(n+1-k)}{(n+1)^2(n+2)}. \tag{4}$$

Thus even a perfectly marginally calibrated method varies across realized calibration samples. This beta statement needs iid sampling, stronger than general finite exchangeability. Given q, a test sample of size m has independent Bernoulli indicators with success probability $F(q)$, so $\mathrm{Var}(\hat{C} \mid q) = F(q)(1-F(q))/m$. An independent finite test sample thus adds binomial noise. For calibration size n, test size m and $\mu = k/(n+1)$, total variance of the test coverage is $v + (\mu(1-\mu) - v)/m$, where v is the beta variance, conditional on the trained continuous score. This follows by total variance and $E[F(q)(1-F(q))] = \mu(1-\mu) - v$.

For coordinatewise Bonferroni calibration, the probability of any missed coordinate is at most $\sum_j \alpha/K = \alpha$. Naive per-coordinate error α only gives the lower bound $\max(0, 1 - K\alpha)$ on joint coverage. Dependence can improve this weak bound but does not supply the desired general guarantee.

A clr difference belongs to the sum-zero subspace. Any orthonormal basis of that subspace preserves its Euclidean norm, giving clr/ilr equivalence. Multiplicative perturbation followed by closure adds a fixed clr vector to both arguments and preserves their difference. These identities justify treating ilr and clr L2 as the same score geometry.

Let c_0, c_1 be group coverages of a fixed calibrated rule and $\pi = \Pr(G = 1)$. Changing only the group mixture to π' changes marginal coverage by

$$C(\pi') - C(\pi) = (\pi' - \pi)(c_1 - c_0). \tag{5}$$

This explains why hidden subgroup imbalance matters under this particular shift. The identity does not cover changes in within-group score laws. Mondrian rank guarantees transfer only because those laws are preserved.

The triangle simplex at $K = 3$ has Euclidean area $\sqrt{3}/2$. A Euclidean ball centered at its barycenter with radius r below its inradius cuts out a disk in its affine plane, giving area fraction $2\pi r^2/\sqrt{3}$. This supplies a known case for checking the uniform-simplex Monte Carlo estimator.

B Executable settings and estimands

Generator and fitted scale. For dimension K, NumPy generator seed $1729 + K$ produces the entries of B independently from $N(0, 0.32^2)$; each row is then centered across components. The same B is retained in all repetitions of that dimension. Let Φ be the training design matrix and $R = \mathrm{diag}(0, 10^{-3}, \ldots, 10^{-3})$. The fitted coefficient matrix is

$$\hat{B} = (\Phi^\top \Phi + R)^{-1} \Phi^\top \mathrm{clr}(Y_{\mathrm{train}}). \tag{6}$$

The scale regression uses targets $t_i = \log \max(\| \mathrm{clr}(Y_i) - \phi(X_i)\hat{B} \|_2^2, 10^{-12})$, coefficient vector $\hat{d} = (\Phi^\top \Phi + R)^{-1} \Phi^\top t$, and

$$\hat{s}(x) = \exp\{\mathrm{clip}(\phi(x)\hat{d}/2, -12, 12)\}. \tag{7}$$

These in-sample residuals can bias the scale level. Calibration absorbs a constant multiplicative error, but does not correct arbitrary errors in the covariate-dependent shape. No true noise scale is

given to the fitted method. Predictor and scale parameters are fixed before calibration; there is no test-based tuning.

Main repetition seeds are 901000 through 901099. Methods share streams within a cell; cells can reuse seed numbers but remain separate estimands. Pilot seeds 900000 and 900001 are excluded from confirmatory inference. The data are newly generated for each full repetition, rather than repeated test draws around one fixed calibration sample. Both environments share the trained model and unweighted thresholds; the oracle rule additionally knows the target-environment density ratio. The 32 area anchors are newly drawn independently of coverage test points in each environment, with 8,192 candidates per anchor.

Area precision. With anchor hit fraction $\hat{v}_a$ and M independent candidates per anchor, the recorded candidate-only standard error is

$$\widehat{\mathrm{SE}}_{\mathrm{MC}} = \sqrt{\frac{A^{-1}\sum_{a=1}^{A}\hat{v}_a(1-\hat{v}_a)}{AM}}. \tag{8}$$

This conditions on the fitted regions and feature anchors. It does not include variation across anchors or across training/calibration datasets. Independent full-repetition intervals do include these sources. Common candidates make method comparisons paired. A finite candidate experiment that returns no hits does not prove zero area.

Mean and paired intervals. For independent repetition estimates $V_1, \ldots, V_R$, the pointwise mean interval is $\bar{V} \pm t_{0.975,R-1}\, s_V/\sqrt{R}$. A paired method contrast first forms $D_r = V_{r,a} - V_{r,b}$ and applies the same formula to D. A paired environment contrast is shifted minus iid within the same trained/calibrated repetition. These are descriptive intervals for mean performance, using a finite-variance normal approximation for the mean; they are not familywise intervals, proof of equivalence, or a guarantee for each realized calibration set. With infinite radii the complete-cell radius mean is omitted and denominators/status are retained; coverage and area retain those full-region outcomes.

The distribution of finite-test coverage estimates, including standard deviation and empirical percentiles, describes repeated-calibration variability. The fraction below nominal minus 0.05 is an empirical tail diagnostic, with a Wilson interval across independent repetition indicators. Its events depend on noisy finite test estimates, so it does not estimate a certified event about exact conditional population coverage.

Rank sensitivity. The focused rank grid is $n \in \{2, 9, 19, 99, 499\}$ and $\alpha \in \{0.05, 0.1, 0.2\}$, with 20,000 independent calibration/test trials per cell and continuous uniform or Bernoulli-0.5 scores. The continuous case checks the rank mean and beta variability; ties are included without randomization. With a Bernoulli score, the conditional covered probability is 0.5 if the threshold is zero and one if it is one or infinite. This makes conservatism directly checkable. Cell seeds follow the recorded rule $310000+$cell index. These score simulations isolate calibration behavior from predictor fit and compositional geometry.

Censoring and replacement. For boundary checks, components of the heterogeneous generator below 0.025 are set to zero and the vector is reclosed. Replacement is the fixed map

$$T_\epsilon(y)_j = \frac{\max(y_j, \epsilon)}{\sum_\ell \max(y_\ell, \epsilon)}. \tag{9}$$

The implementation uses this clipping-and-closure map, not an additive pseudocount. Each $K \in \{3, 10\}$ uses 50 fresh seeds 902000 through 902049, $n = 499$, $\alpha = 0.1$ and 5,000 iid test observations. Original datasets are identical across the three replacement values. The center and scale are trained on transformed responses. Original-target raw methods calibrate their raw residual scores on censored Y and test on censored Y. Transformed-target methods consistently calibrate and test on $T_\epsilon(Y)$. An additional original-target clr diagnostic uses the transformed-calibrated rule: finite clr regions reject original responses with any zero and score only positive responses; an infinite threshold invokes the explicit whole-simplex fallback.

The transformed event $T_\epsilon(Y) \in C_\epsilon(X)$ can equivalently define an original-space inverse-image region $\{y : T_\epsilon(y) \in C_\epsilon(X)\}$. Original-response membership in this in-

verse image equals the transformed membership observation by observation, so its measured coverage and interval equal the transformed estimates, including at zeros; invertibility is unnecessary. This is an exact reinterpretation of existing events, recorded in `results/boundary_summary/pullback_equivalence.csv`, not a new simulation. It is distinct from directly testing $Y \in C_\epsilon(X)$. The reported direct original clr diagnostic uses the latter rule, and inherits no transformed-target guarantee. No original-simplex area of the inverse image was estimated, so its efficiency is unresolved.

C Research workflow and reproducibility scope

The research workflow uses language-model agents for literature assessment, study planning, implementation, analysis, presentation and manuscript integration, with separate review functions. This is substantive assistance to the research process, beyond formatting alone. The numerical predictor and data generator are the explicitly specified regression and synthetic sampling procedures; they do not call a language model or a paid prediction API. Agent-generated code and reasoning require the same executable checks, evidence provenance and independent review as other research work. No agent identity is presented as a human author.

The supplied task restricts access to a withheld prior study and its derivatives. Background reading is deliberately bounded to allowed starter sources and directly unrelated official venue instructions, with no attempt to identify the withheld work. The reading provenance and consulted versions accompany the LaTeX source. Original sources that could not be obtained are not cited merely because they appear in the supplied list. This restriction limits an exhaustive novelty assessment and is not evidence that no other related work exists.

D Reproduction and resource record

The accompanying repository supplies all numerical and presentation code, a synthetic generator instead of a downloaded dataset, raw task-run outputs, summary tables, manifests, source hashes, exact commands and a figure/table inventory. The analysis environment records Python 3.11.2, NumPy 2.4.6, SciPy 1.17.1 and pandas 3.0.6. The rendering runtime is recorded with its input hashes. The official NeurIPS 2026 style is unmodified; the root PDF is rebuilt from the delivered LaTeX. Inline estimates are extracted by a paper-owned script, with exact file hashes, selection keys and columns in the accompanying claim-provenance record.

Runs were CPU-only in resource containers with ceilings of two CPUs and 4 GiB memory, on a shared host; the documented commands set BLAS/OpenMP to one thread. Core running wall/CPU times were 116.110/124.165104 seconds and the oracle/rank/boundary extension used 43.878/45.080784 seconds, excluding startup. The TV/KL supplement used 45.631351 CPU seconds and resource-reported running time 44.498 seconds; its per-cell wall-time sum is 59.927 seconds, a separately recorded timing measure. The implementation pilot took approximately three seconds; it and the separate score-implementation pilot are excluded from the empirical claims. No resource run failed and no final repetition was excluded. Peak memory was not measured. Exact per-cell times are retained, rather than treating these timings as hardware-independent promises. Reproduction requires a few minutes plus summarization on a comparable host, with no GPU or paid prediction API.

No external dataset or pretrained model was used. Existing dependency creators, versions and installed terms are listed in the accompanying asset inventory. Official venue-source hashes and their acquisition record are retained. No blanket license for all archived conference assets or general license grant for newly generated code/data is invented. Sources not obtained as originals are not represented as consulted originals. The supplied source-access logs report no prohibited-study encounter; the bounded permitted-source search is not an exhaustive novelty claim.

E Supporting displays and full-cell estimates

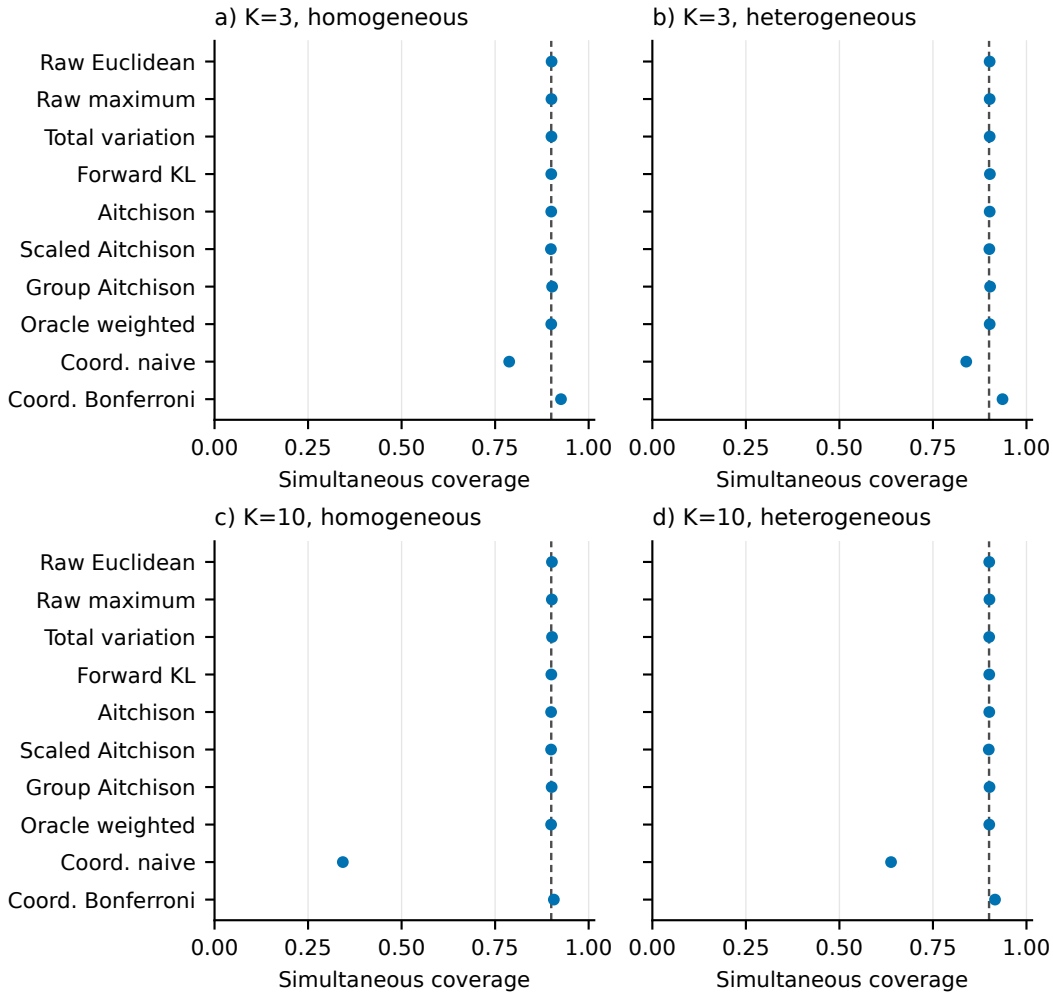


Figure 3: **Joint response coverage under iid sampling.** Dots are means across 100 independent train/calibration/test replications; bars are pointwise 95% t intervals for these means, not simultaneous bands or intervals for a single calibrated run. Each replication uses 1000 training and 499 calibration observations and 5000 test observations per environment. The dashed line marks nominal $1 - \alpha = 0.90$. The target is the entire observed composition, not a class label. "Coord. naive" calibrates each coordinate at α and is a negative control; "Coord. Bonferroni" uses α/K. Raw Euclidean and maximum regions are intersected with the simplex; Aitchison denotes the Euclidean norm of clr differences. The oracle weighted method uses unit weights under iid sampling. TV is exactly raw maximum at $K = 3$ and is distinct at $K = 10$; KL is forward $D_{\mathrm{KL}}(y\|\hat{\mu})$.

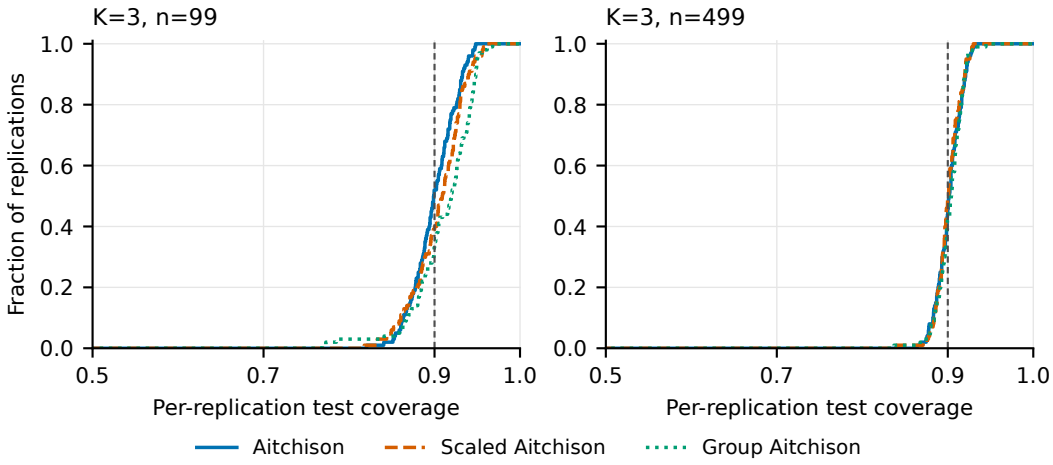


Figure 4: **Mean validity does not guarantee every calibrated run.** Empirical cumulative distributions of finite-test joint coverage across 100 independent datasets in the iid heterogeneous $K = 3$ setting, for calibration sizes 99 and 499. Global, training-only scaled and observed-group Aitchison methods share datasets within each replication. Each plotted coverage uses 5000 independent test responses conditional on its fitted and calibrated predictor; it still contains finite-test noise. The dashed line marks 0.90. These distributions and their empirical lower tails are descriptive, not certified probabilities of population conditional validity.

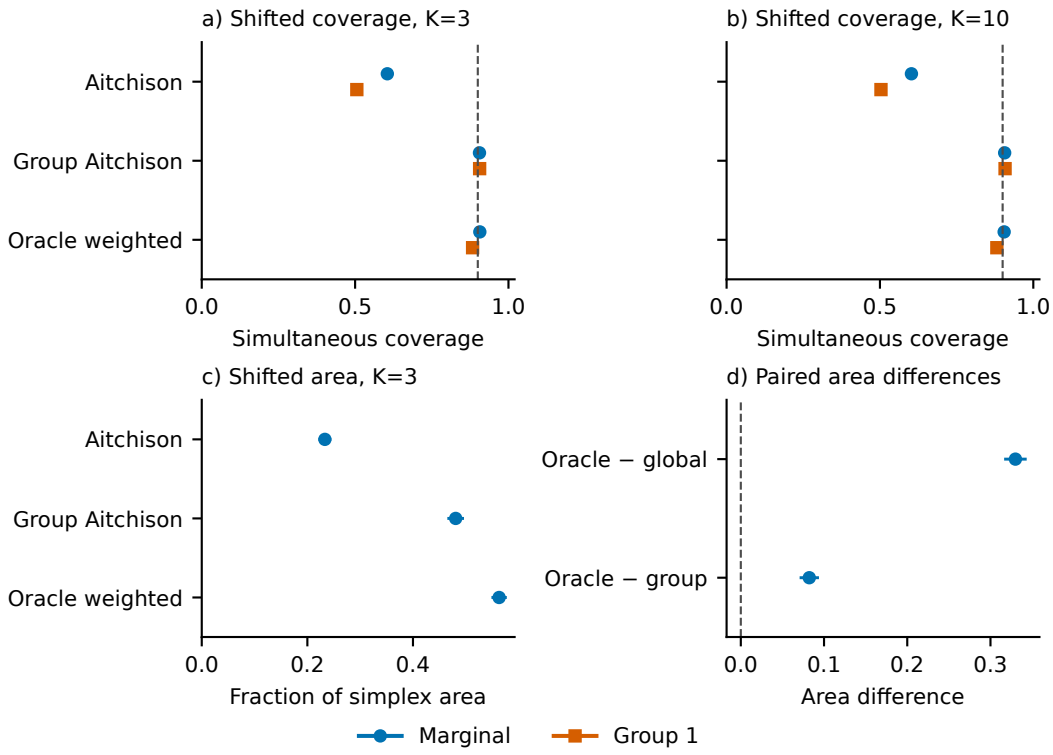


Figure 5: **Known-ratio weighting under the controlled mixture shift.** Heterogeneous mechanism, calibration size 499; test group-1 probability changes from 0.2 to 0.8 with within-group laws fixed. Panels (a,b) show shifted marginal and higher-noise-group joint coverage; (c) shows shifted-feature-averaged Euclidean simplex-area fractions at $K = 3$; (d) shows within-seed paired oracle-minus-global and oracle-minus-group area differences. Dots/bars are means/pointwise 95% t intervals across independent dataset replications, not simultaneous bands. Oracle weights use exact generator density ratios with test weight at infinity. This does not evaluate learned ratios or arbitrary-shift robustness. Area needs the coverage context in (a,b).

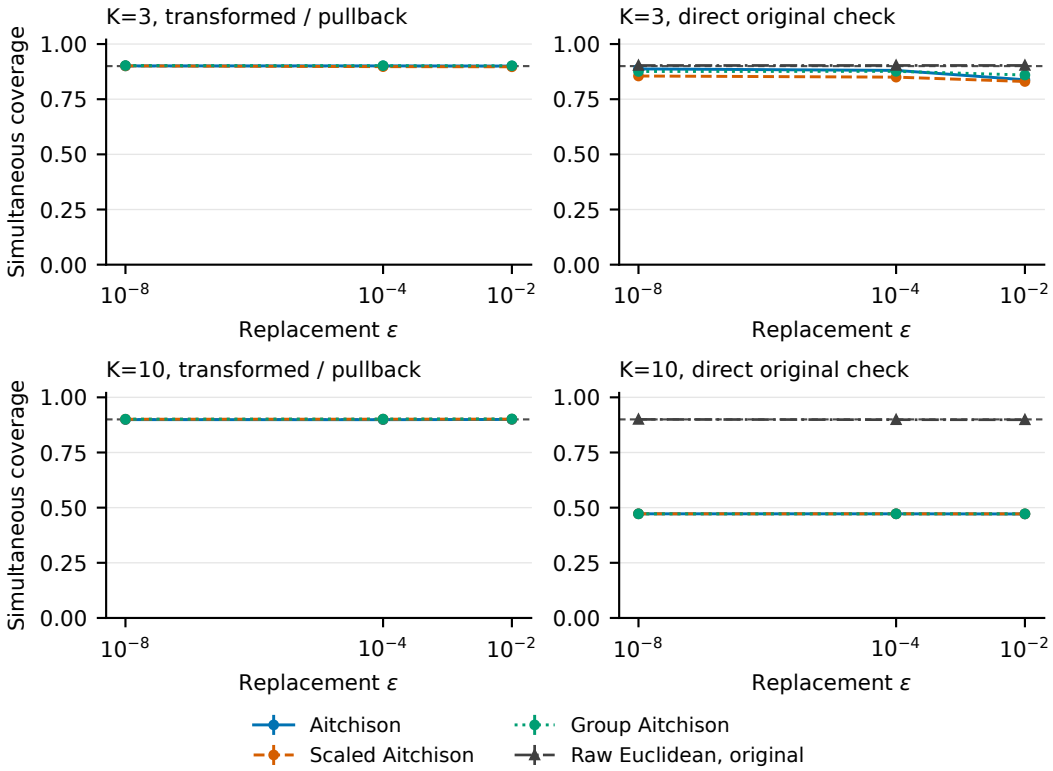


Figure 6: **Replacement changes the coverage target.** Exact zeros arise by censoring components below 0.025 and reclosing. The fixed transform is $T_\epsilon(y)_j = \max(y_j, \epsilon)/\sum_l \max(y_l, \epsilon)$. Left: clr methods calibrate and test on $T_\epsilon(Y)$. These same values and intervals exactly equal original-response coverage of the inverse-image region $\{y : T_\epsilon(y) \in C_\epsilon(x)\}$ by event identity; this is not another simulation. Right: those same finite clr regions are tested against original Y, rejecting boundary points, while the raw Euclidean reference separately calibrates and tests on original Y. All fits use transformed training responses. Dots are means and bars pointwise 95% t intervals across 50 independent datasets, with 1000 training, 499 calibration and 5000 iid test observations each. The same original datasets are reused across replacement values. The dashed line marks 0.90; the original-clr check has no nominal guarantee. The right-panel diagnostic is a different region from the left-panel inverse image. Replacement does not inherently prevent coverage of original zeros. No inverse-image area was estimated, and this censoring test does not establish structural-zero performance.

Table 2: **Finite-test lower tails across calibrated datasets.** Heterogeneous iid cells; 100 independent replications each. The coverage 10th percentile is descriptive; the fraction of replications with 5000-test-point coverage below .85 has a pointwise 95% Wilson interval. These are not certified population conditional-coverage probabilities because finite-test noise remains. Full diagnostics and infinite-region counts are in `tables/calibration_diagnostics.csv`.

K	n	Method	Coverage 10th percentile	Fraction below .85 [95% CI]
3	99	Aitchison	0.864	0.02 [0.01, 0.07]
3	99	Group Aitchison	0.869	0.04 [0.02, 0.10]
3	99	Scaled Aitchison	0.861	0.06 [0.03, 0.12]
3	499	Aitchison	0.883	0.00 [0.00, 0.04]
3	499	Group Aitchison	0.884	0.01 [0.00, 0.05]
3	499	Scaled Aitchison	0.884	0.01 [0.00, 0.05]
10	99	Aitchison	0.858	0.05 [0.02, 0.11]
10	99	Group Aitchison	0.871	0.05 [0.02, 0.11]
10	99	Scaled Aitchison	0.867	0.06 [0.03, 0.12]
10	499	Aitchison	0.884	0.00 [0.00, 0.04]
10	499	Group Aitchison	0.882	0.00 [0.00, 0.04]
10	499	Scaled Aitchison	0.879	0.00 [0.00, 0.04]

Table 3: **Source-driven score controls under a shared fitted predictor.** $K = 3$, calibration size 499; paired within-seed area differences with pointwise 95% t intervals over 100 independent datasets. Negative values favor the first score only for the chosen uniform Euclidean simplex-area measure; judge them with subgroup and shifted coverage, not alone. Intervals including zero leave the ordering unresolved. TV is exactly raw maximum at $K = 3$, including the zero paired differences shown here; these are not independent geometries. KL is forward $D_{\mathrm{KL}}(y\|\hat{\mu})$. These controls were requested from permitted prior work after core results were inspected, before their own outcomes were known; they compare scores, not the prior study full fitted pipeline. Candidate-resolution diagnostics and all other cells are in `tables/area_precision.csv` and `tables/paired_method_estimates.csv`.

Mechanism	Area difference	IID [95% CI]	Shifted [95% CI]
homogeneous	TV − Euclidean	-0.0027 [-0.0041, -0.0013]	-0.0025 [-0.0038, -0.0011]
homogeneous	TV − Maximum	0.0000 [0.0000, 0.0000]	0.0000 [0.0000, 0.0000]
homogeneous	TV − Aitchison	0.0576 [0.0544, 0.0608]	0.0620 [0.0589, 0.0651]
homogeneous	KL − Euclidean	-0.0476 [-0.0493, -0.0459]	-0.0482 [-0.0498, -0.0465]
homogeneous	KL − Maximum	-0.0448 [-0.0466, -0.0431]	-0.0457 [-0.0474, -0.0439]
homogeneous	KL − Aitchison	0.0128 [0.0104, 0.0151]	0.0163 [0.0142, 0.0185]
heterogeneous	TV − Euclidean	-0.0013 [-0.0031, 0.0005]	-0.0010 [-0.0027, 0.0007]
heterogeneous	TV − Maximum	0.0000 [0.0000, 0.0000]	0.0000 [0.0000, 0.0000]
heterogeneous	TV − Aitchison	-0.0222 [-0.0296, -0.0148]	-0.0125 [-0.0190, -0.0060]
heterogeneous	KL − Euclidean	-0.0158 [-0.0197, -0.0120]	-0.0184 [-0.0220, -0.0147]
heterogeneous	KL − Maximum	-0.0145 [-0.0181, -0.0109]	-0.0174 [-0.0208, -0.0139]
heterogeneous	KL − Aitchison	-0.0367 [-0.0423, -0.0312]	-0.0298 [-0.0347, -0.0250]

Finite ranks and ties. Each row uses 20,000 independent trials. Rank is $\lceil (n+1)(1-\alpha) \rceil$; ∞ denotes the whole-domain fallback. Theory is exact expected coverage including conservative Bernoulli ties. Test intervals are 95% Wilson; calibration coverage (CDF at the threshold) has a pointwise 95% t interval for its trial mean. Variance checks and seeds remain in `tables/rank_checks.csv`.

n	α	Scores	Rank	Theory	Test [95% CI]	Calibration [95% CI]
2	0.05	bernoulli	∞	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
9	0.05	bernoulli	∞	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
19	0.05	bernoulli	19	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
99	0.05	bernoulli	95	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
499	0.05	bernoulli	475	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
2	0.1	bernoulli	∞	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
9	0.1	bernoulli	9	0.999	0.9991 [0.9986, 0.9995]	0.999 [0.999, 0.999]
19	0.1	bernoulli	18	1.000	1.0000 [0.9997, 1.0000]	1.000 [1.000, 1.000]
99	0.1	bernoulli	90	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
499	0.1	bernoulli	450	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
2	0.2	bernoulli	∞	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
9	0.2	bernoulli	8	0.990	0.9896 [0.9881, 0.9909]	0.990 [0.989, 0.991]
19	0.2	bernoulli	16	0.999	0.9990 [0.9985, 0.9994]	0.999 [0.999, 0.999]
99	0.2	bernoulli	80	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
499	0.2	bernoulli	400	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
2	0.05	uniform	∞	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
9	0.05	uniform	∞	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
19	0.05	uniform	19	0.950	0.9511 [0.9480, 0.9540]	0.950 [0.949, 0.951]
99	0.05	uniform	95	0.950	0.9489 [0.9458, 0.9519]	0.950 [0.950, 0.950]
499	0.05	uniform	475	0.950	0.9492 [0.9460, 0.9521]	0.950 [0.950, 0.950]
2	0.1	uniform	∞	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
9	0.1	uniform	9	0.900	0.8979 [0.8936, 0.9020]	0.899 [0.898, 0.900]
19	0.1	uniform	18	0.900	0.9002 [0.8959, 0.9042]	0.900 [0.899, 0.901]
99	0.1	uniform	90	0.900	0.8976 [0.8934, 0.9018]	0.900 [0.899, 0.900]
499	0.1	uniform	450	0.900	0.9001 [0.8959, 0.9042]	0.900 [0.900, 0.900]
2	0.2	uniform	∞	1.000	1.0000 [0.9998, 1.0000]	1.000 [1.000, 1.000]
9	0.2	uniform	8	0.800	0.8023 [0.7967, 0.8077]	0.801 [0.799, 0.803]
19	0.2	uniform	16	0.800	0.7976 [0.7920, 0.8031]	0.800 [0.799, 0.802]
99	0.2	uniform	80	0.800	0.8014 [0.7959, 0.8069]	0.800 [0.800, 0.801]
499	0.2	uniform	400	0.800	0.7974 [0.7918, 0.8030]	0.800 [0.800, 0.800]

E.1 All main-grid numeric panels

The following panels retain all main dimension, calibration, noise and environment cells. Numerical source files also retain group 0 values, raw replication estimates and paired contrasts. Pointwise intervals do not imply simultaneous coverage guarantees across panels.

Table 5: **Full-cell estimates: $K = 10$, calibration size 499, heterogeneous.** IID test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.900 [0.898, 0.903]	0.502 [0.489, 0.515]	–
Raw maximum	0.901 [0.898, 0.904]	0.507 [0.494, 0.520]	–
Total variation	0.900 [0.898, 0.903]	0.500 [0.487, 0.514]	–
Forward KL	0.901 [0.898, 0.903]	0.502 [0.489, 0.514]	–
Aitchison	0.901 [0.898, 0.903]	0.502 [0.489, 0.514]	–
Scaled Aitchison	0.899 [0.896, 0.902]	0.894 [0.889, 0.899]	–
Group Aitchison	0.901 [0.898, 0.904]	0.907 [0.901, 0.913]	–
Oracle weighted	0.901 [0.898, 0.903]	0.502 [0.489, 0.514]	–
Coord. naive	0.638 [0.633, 0.643]	0.011 [0.010, 0.012]	–
Coord. Bonferroni	0.916 [0.914, 0.918]	0.582 [0.570, 0.593]	–

Table 6: **Full-cell estimates: $K = 10$, calibration size 499, heterogeneous.** Controlled mixture-shift test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.601 [0.591, 0.611]	0.501 [0.488, 0.514]	–
Raw maximum	0.605 [0.595, 0.615]	0.507 [0.494, 0.519]	–
Total variation	0.601 [0.590, 0.611]	0.501 [0.488, 0.513]	–
Forward KL	0.602 [0.592, 0.612]	0.502 [0.490, 0.515]	–
Aitchison	0.603 [0.593, 0.612]	0.503 [0.492, 0.515]	–
Scaled Aitchison	0.896 [0.892, 0.900]	0.895 [0.890, 0.899]	–
Group Aitchison	0.907 [0.902, 0.911]	0.908 [0.902, 0.914]	–
Oracle weighted	0.905 [0.899, 0.911]	0.881 [0.874, 0.888]	–
Coord. naive	0.168 [0.167, 0.170]	0.011 [0.011, 0.012]	–
Coord. Bonferroni	0.664 [0.655, 0.672]	0.580 [0.568, 0.591]	–

Table 7: **Full-cell estimates: $K = 10$, calibration size 499, homogeneous.** IID test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.902 [0.899, 0.905]	0.906 [0.903, 0.909]	–
Raw maximum	0.902 [0.899, 0.904]	0.904 [0.901, 0.907]	–
Total variation	0.902 [0.899, 0.905]	0.908 [0.905, 0.912]	–
Forward KL	0.900 [0.897, 0.903]	0.906 [0.903, 0.909]	–
Aitchison	0.900 [0.897, 0.903]	0.898 [0.894, 0.901]	–
Scaled Aitchison	0.900 [0.896, 0.903]	0.895 [0.890, 0.899]	–
Group Aitchison	0.901 [0.898, 0.904]	0.903 [0.897, 0.910]	–
Oracle weighted	0.900 [0.897, 0.903]	0.898 [0.894, 0.901]	–
Coord. naive	0.343 [0.339, 0.347]	0.318 [0.313, 0.322]	–
Coord. Bonferroni	0.907 [0.904, 0.910]	0.893 [0.889, 0.897]	–

Table 8: **Full-cell estimates: $K = 10$, calibration size 499, homogeneous.** Controlled mixture-shift test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.904 [0.901, 0.906]	0.904 [0.902, 0.907]	–
Raw maximum	0.902 [0.899, 0.905]	0.902 [0.899, 0.905]	–
Total variation	0.907 [0.904, 0.910]	0.908 [0.906, 0.911]	–
Forward KL	0.905 [0.902, 0.908]	0.907 [0.904, 0.910]	–
Aitchison	0.899 [0.896, 0.902]	0.899 [0.896, 0.902]	–
Scaled Aitchison	0.897 [0.892, 0.901]	0.896 [0.891, 0.900]	–
Group Aitchison	0.903 [0.898, 0.909]	0.904 [0.897, 0.911]	–
Oracle weighted	0.903 [0.898, 0.908]	0.904 [0.899, 0.909]	–
Coord. naive	0.325 [0.321, 0.329]	0.319 [0.315, 0.323]	–
Coord. Bonferroni	0.897 [0.894, 0.900]	0.894 [0.890, 0.897]	–

Table 9: **Full-cell estimates: $K = 10$, calibration size 99, heterogeneous.** IID test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.898 [0.892, 0.904]	0.492 [0.462, 0.522]	–
Raw maximum	0.898 [0.892, 0.905]	0.499 [0.470, 0.528]	–
Total variation	0.897 [0.891, 0.903]	0.484 [0.454, 0.515]	–
Forward KL	0.898 [0.892, 0.904]	0.490 [0.460, 0.520]	–
Aitchison	0.899 [0.893, 0.905]	0.494 [0.463, 0.525]	–
Scaled Aitchison	0.899 [0.894, 0.905]	0.893 [0.886, 0.900]	–
Group Aitchison	0.908 [0.903, 0.914]	0.912 [0.898, 0.925]	–
Oracle weighted	0.899 [0.893, 0.905]	0.494 [0.463, 0.525]	–
Coord. naive	0.621 [0.610, 0.633]	0.012 [0.011, 0.014]	–
Coord. Bonferroni	0.917 [0.912, 0.922]	0.593 [0.573, 0.614]	–

Table 10: **Full-cell estimates: $K = 10$, calibration size 99, heterogeneous.** Controlled mixture-shift test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.593 [0.569, 0.618]	0.492 [0.462, 0.522]	–
Raw maximum	0.599 [0.576, 0.623]	0.500 [0.471, 0.529]	–
Total variation	0.588 [0.564, 0.613]	0.486 [0.455, 0.516]	–
Forward KL	0.593 [0.569, 0.617]	0.491 [0.462, 0.521]	–
Aitchison	0.595 [0.570, 0.619]	0.493 [0.463, 0.523]	–
Scaled Aitchison	0.895 [0.889, 0.902]	0.894 [0.888, 0.901]	–
Group Aitchison	0.912 [0.901, 0.923]	0.913 [0.900, 0.927]	–
Oracle weighted	0.912 [0.900, 0.924]	0.890 [0.875, 0.905]	–
Coord. naive	0.165 [0.161, 0.169]	0.012 [0.011, 0.013]	–
Coord. Bonferroni	0.673 [0.656, 0.689]	0.592 [0.572, 0.612]	–

Table 11: **Full-cell estimates: $K = 10$, calibration size 99, homogeneous.** IID test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.907 [0.901, 0.912]	0.910 [0.905, 0.915]	–
Raw maximum	0.908 [0.902, 0.913]	0.909 [0.904, 0.914]	–
Total variation	0.905 [0.900, 0.911]	0.911 [0.906, 0.917]	–
Forward KL	0.903 [0.897, 0.908]	0.909 [0.903, 0.914]	–
Aitchison	0.899 [0.893, 0.905]	0.897 [0.891, 0.902]	–
Scaled Aitchison	0.898 [0.893, 0.904]	0.893 [0.886, 0.899]	–
Group Aitchison	0.908 [0.903, 0.913]	0.915 [0.902, 0.927]	–
Oracle weighted	0.899 [0.893, 0.905]	0.897 [0.891, 0.902]	–
Coord. naive	0.343 [0.336, 0.350]	0.322 [0.315, 0.329]	–
Coord. Bonferroni	0.910 [0.904, 0.915]	0.896 [0.889, 0.903]	–

Table 12: **Full-cell estimates: $K = 10$, calibration size 99, homogeneous.** Controlled mixture-shift test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.908 [0.903, 0.914]	0.909 [0.904, 0.914]	–
Raw maximum	0.908 [0.903, 0.913]	0.908 [0.903, 0.913]	–
Total variation	0.909 [0.904, 0.914]	0.911 [0.906, 0.916]	–
Forward KL	0.907 [0.901, 0.912]	0.908 [0.903, 0.913]	–
Aitchison	0.898 [0.893, 0.904]	0.899 [0.893, 0.905]	–
Scaled Aitchison	0.895 [0.889, 0.901]	0.894 [0.888, 0.901]	–
Group Aitchison	0.912 [0.902, 0.922]	0.914 [0.902, 0.927]	–
Oracle weighted	0.911 [0.902, 0.920]	0.917 [0.908, 0.926]	–
Coord. naive	0.327 [0.320, 0.333]	0.320 [0.314, 0.327]	–
Coord. Bonferroni	0.899 [0.893, 0.906]	0.896 [0.889, 0.903]	–

Table 13: **Full-cell estimates: $K = 3$, calibration size 499, heterogeneous.** IID test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.902 [0.899, 0.904]	0.512 [0.499, 0.525]	0.232 [0.222, 0.241]
Raw maximum	0.902 [0.899, 0.905]	0.512 [0.499, 0.525]	0.230 [0.221, 0.240]
Total variation	0.902 [0.899, 0.905]	0.512 [0.499, 0.525]	0.230 [0.221, 0.240]
Forward KL	0.902 [0.900, 0.905]	0.511 [0.499, 0.523]	0.216 [0.208, 0.224]
Aitchison	0.902 [0.899, 0.905]	0.507 [0.493, 0.521]	0.253 [0.243, 0.262]
Scaled Aitchison	0.901 [0.898, 0.904]	0.899 [0.894, 0.904]	0.157 [0.150, 0.165]
Group Aitchison	0.903 [0.900, 0.906]	0.906 [0.901, 0.912]	0.164 [0.155, 0.172]
Oracle weighted	0.902 [0.899, 0.905]	0.507 [0.493, 0.521]	0.253 [0.243, 0.262]
Coord. naive	0.839 [0.836, 0.842]	0.324 [0.317, 0.331]	0.090 [0.087, 0.092]
Coord. Bonferroni	0.936 [0.934, 0.938]	0.681 [0.671, 0.690]	0.317 [0.308, 0.326]

Table 14: **Full-cell estimates: $K = 3$, calibration size 499, heterogeneous.** Controlled mixture-shift test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.608 [0.598, 0.618]	0.510 [0.497, 0.522]	0.222 [0.212, 0.231]
Raw maximum	0.609 [0.598, 0.619]	0.510 [0.498, 0.523]	0.221 [0.211, 0.231]
Total variation	0.609 [0.598, 0.619]	0.510 [0.498, 0.523]	0.221 [0.211, 0.231]
Forward KL	0.608 [0.599, 0.618]	0.510 [0.498, 0.522]	0.203 [0.195, 0.212]
Aitchison	0.605 [0.594, 0.616]	0.506 [0.492, 0.519]	0.233 [0.224, 0.243]
Scaled Aitchison	0.898 [0.893, 0.902]	0.896 [0.891, 0.901]	0.462 [0.450, 0.474]
Group Aitchison	0.905 [0.901, 0.910]	0.906 [0.901, 0.911]	0.481 [0.468, 0.494]
Oracle weighted	0.907 [0.902, 0.911]	0.883 [0.877, 0.889]	0.563 [0.551, 0.575]
Coord. naive	0.451 [0.445, 0.456]	0.320 [0.314, 0.327]	0.089 [0.086, 0.091]
Coord. Bonferroni	0.744 [0.736, 0.751]	0.679 [0.669, 0.689]	0.308 [0.299, 0.317]

Table 15: **Full-cell estimates: $K = 3$, calibration size 499, homogeneous.** IID test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.901 [0.898, 0.904]	0.895 [0.891, 0.898]	0.266 [0.261, 0.270]
Raw maximum	0.901 [0.897, 0.904]	0.898 [0.894, 0.901]	0.263 [0.259, 0.268]
Total variation	0.901 [0.897, 0.904]	0.898 [0.894, 0.901]	0.263 [0.259, 0.268]
Forward KL	0.900 [0.897, 0.903]	0.899 [0.896, 0.902]	0.218 [0.214, 0.222]
Aitchison	0.900 [0.897, 0.903]	0.901 [0.898, 0.905]	0.205 [0.201, 0.209]
Scaled Aitchison	0.899 [0.896, 0.902]	0.896 [0.891, 0.902]	0.207 [0.203, 0.211]
Group Aitchison	0.902 [0.899, 0.905]	0.907 [0.902, 0.912]	0.208 [0.204, 0.212]
Oracle weighted	0.900 [0.897, 0.903]	0.901 [0.898, 0.905]	0.205 [0.201, 0.209]
Coord. naive	0.788 [0.784, 0.791]	0.805 [0.800, 0.809]	0.153 [0.151, 0.155]
Coord. Bonferroni	0.926 [0.923, 0.929]	0.931 [0.928, 0.934]	0.272 [0.267, 0.276]

Table 16: **Full-cell estimates: $K = 3$, calibration size 499, homogeneous.** Controlled mixture-shift test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.896 [0.892, 0.899]	0.894 [0.891, 0.897]	0.254 [0.249, 0.258]
Raw maximum	0.898 [0.895, 0.901]	0.897 [0.894, 0.900]	0.251 [0.247, 0.255]
Total variation	0.898 [0.895, 0.901]	0.897 [0.894, 0.900]	0.251 [0.247, 0.255]
Forward KL	0.899 [0.896, 0.902]	0.898 [0.895, 0.901]	0.206 [0.202, 0.209]
Aitchison	0.900 [0.897, 0.903]	0.899 [0.896, 0.902]	0.189 [0.186, 0.193]
Scaled Aitchison	0.896 [0.892, 0.900]	0.895 [0.890, 0.900]	0.190 [0.185, 0.194]
Group Aitchison	0.905 [0.901, 0.909]	0.905 [0.900, 0.911]	0.195 [0.190, 0.200]
Oracle weighted	0.904 [0.899, 0.908]	0.905 [0.900, 0.909]	0.194 [0.189, 0.198]
Coord. naive	0.798 [0.795, 0.802]	0.802 [0.798, 0.806]	0.149 [0.147, 0.151]
Coord. Bonferroni	0.930 [0.927, 0.933]	0.931 [0.928, 0.934]	0.262 [0.258, 0.266]

Table 17: **Full-cell estimates: $K = 3$, calibration size 99, heterogeneous.** IID test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.900 [0.894, 0.905]	0.510 [0.487, 0.534]	0.237 [0.218, 0.255]
Raw maximum	0.900 [0.894, 0.905]	0.511 [0.488, 0.534]	0.235 [0.217, 0.253]
Total variation	0.900 [0.894, 0.905]	0.511 [0.488, 0.534]	0.235 [0.217, 0.253]
Forward KL	0.900 [0.895, 0.906]	0.506 [0.481, 0.532]	0.220 [0.203, 0.237]
Aitchison	0.900 [0.894, 0.905]	0.498 [0.473, 0.523]	0.250 [0.234, 0.266]
Scaled Aitchison	0.904 [0.898, 0.910]	0.901 [0.894, 0.908]	0.149 [0.139, 0.159]
Group Aitchison	0.911 [0.904, 0.918]	0.924 [0.913, 0.935]	0.164 [0.153, 0.175]
Oracle weighted	0.900 [0.894, 0.905]	0.498 [0.473, 0.523]	0.250 [0.234, 0.266]
Coord. naive	0.833 [0.825, 0.840]	0.328 [0.313, 0.344]	0.095 [0.088, 0.102]
Coord. Bonferroni	0.939 [0.935, 0.942]	0.696 [0.678, 0.713]	0.341 [0.324, 0.357]

Table 18: **Full-cell estimates: $K = 3$, calibration size 99, heterogeneous.** Controlled mixture-shift test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.608 [0.589, 0.627]	0.510 [0.487, 0.534]	0.226 [0.209, 0.244]
Raw maximum	0.608 [0.590, 0.627]	0.511 [0.488, 0.534]	0.225 [0.208, 0.242]
Total variation	0.608 [0.590, 0.627]	0.511 [0.488, 0.534]	0.225 [0.208, 0.242]
Forward KL	0.605 [0.584, 0.626]	0.506 [0.481, 0.532]	0.207 [0.190, 0.223]
Aitchison	0.598 [0.578, 0.618]	0.498 [0.473, 0.523]	0.232 [0.217, 0.247]
Scaled Aitchison	0.902 [0.896, 0.909]	0.901 [0.894, 0.908]	0.470 [0.456, 0.483]
Group Aitchison	0.921 [0.912, 0.930]	0.924 [0.913, 0.935]	0.532 [0.510, 0.554]
Oracle weighted	0.920 [0.911, 0.930]	0.900 [0.889, 0.912]	0.595 [0.575, 0.616]
Coord. naive	0.455 [0.442, 0.468]	0.328 [0.313, 0.343]	0.094 [0.087, 0.101]
Coord. Bonferroni	0.755 [0.741, 0.769]	0.694 [0.676, 0.711]	0.333 [0.316, 0.349]

Table 19: **Full-cell estimates: $K = 3$, calibration size 99, homogeneous.** IID test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.903 [0.897, 0.909]	0.895 [0.889, 0.902]	0.273 [0.265, 0.282]
Raw maximum	0.904 [0.897, 0.910]	0.899 [0.892, 0.906]	0.271 [0.263, 0.280]
Total variation	0.904 [0.897, 0.910]	0.899 [0.892, 0.906]	0.271 [0.263, 0.280]
Forward KL	0.903 [0.897, 0.910]	0.901 [0.894, 0.907]	0.225 [0.218, 0.232]
Aitchison	0.901 [0.895, 0.907]	0.900 [0.894, 0.907]	0.209 [0.202, 0.215]
Scaled Aitchison	0.902 [0.896, 0.908]	0.899 [0.891, 0.906]	0.212 [0.205, 0.219]
Group Aitchison	0.912 [0.906, 0.918]	0.926 [0.915, 0.937]	0.224 [0.217, 0.231]
Oracle weighted	0.901 [0.895, 0.907]	0.900 [0.894, 0.907]	0.209 [0.202, 0.215]
Coord. naive	0.791 [0.783, 0.799]	0.806 [0.798, 0.814]	0.157 [0.152, 0.161]
Coord. Bonferroni	0.931 [0.926, 0.937]	0.935 [0.930, 0.941]	0.289 [0.280, 0.299]

Table 20: **Full-cell estimates: $K = 3$, calibration size 99, homogeneous.** Controlled mixture-shift test environment. All methods share 100 independent datasets. Brackets are pointwise 95% t intervals for replication means, not simultaneous bands. Area is the common Euclidean simplex fraction at $K = 3$; – means unmeasured at $K = 10$. TV duplicates raw maximum at $K = 3$; forward KL uses $y\|\hat{\mu}$. Group 1 is the prespecified higher-noise group in the heterogeneous mechanism. Full group-0 estimates and denominators are in `tables/all_estimates.csv`.

Method	Joint coverage	Group 1 coverage	Area fraction
Raw Euclidean	0.898 [0.892, 0.905]	0.896 [0.890, 0.902]	0.262 [0.253, 0.270]
Raw maximum	0.901 [0.895, 0.908]	0.900 [0.893, 0.906]	0.260 [0.252, 0.269]
Total variation	0.901 [0.895, 0.908]	0.900 [0.893, 0.906]	0.260 [0.252, 0.269]
Forward KL	0.902 [0.896, 0.909]	0.901 [0.895, 0.907]	0.212 [0.205, 0.220]
Aitchison	0.901 [0.895, 0.907]	0.900 [0.894, 0.906]	0.193 [0.187, 0.200]
Scaled Aitchison	0.900 [0.893, 0.907]	0.899 [0.892, 0.906]	0.195 [0.188, 0.202]
Group Aitchison	0.924 [0.915, 0.932]	0.927 [0.916, 0.938]	0.233 [0.219, 0.247]
Oracle weighted	0.918 [0.909, 0.927]	0.923 [0.914, 0.933]	0.218 [0.207, 0.228]
Coord. naive	0.802 [0.794, 0.809]	0.805 [0.797, 0.813]	0.152 [0.148, 0.157]
Coord. Bonferroni	0.935 [0.930, 0.941]	0.936 [0.931, 0.941]	0.280 [0.271, 0.288]

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